TD5: WORKSHOP LAYOUT

Exercise 1:

You have to set up a manufacturing workshop with 7 stations: A, B, C, D, E, F, G.

The family of parts manufactured by this workshop is composed of 5 parts: P1, P2, P3, P4, P5.

The following table summarizes the manufacturing routings of these five parts as well as the number of transfers for each part.

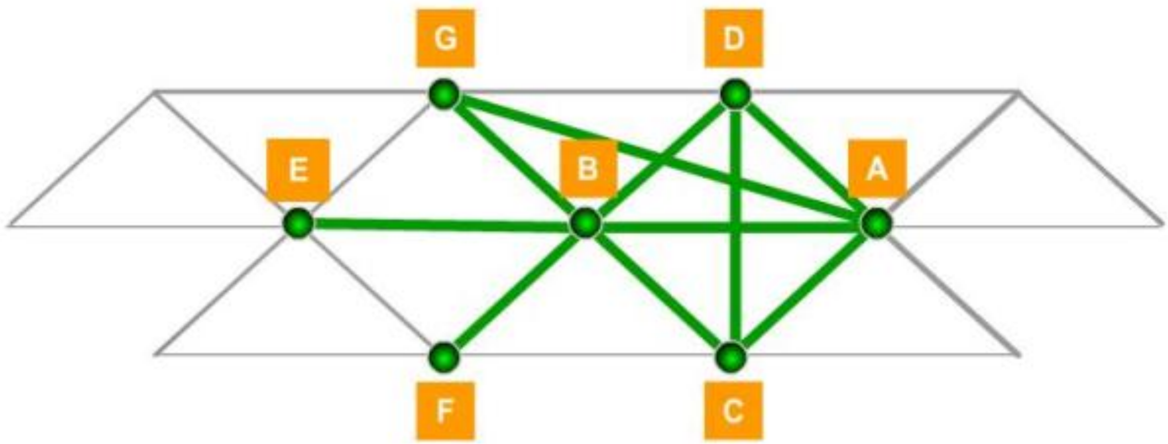
Parts	Routings						Nb of transfer
	10	20	30	40	50	60	
P1	A	D	B	E			25
P2	F	B	D	A	G	B	43
P3	F	B	D	A			15
P4	A	C	B				24
P5	A	B	C	D			90

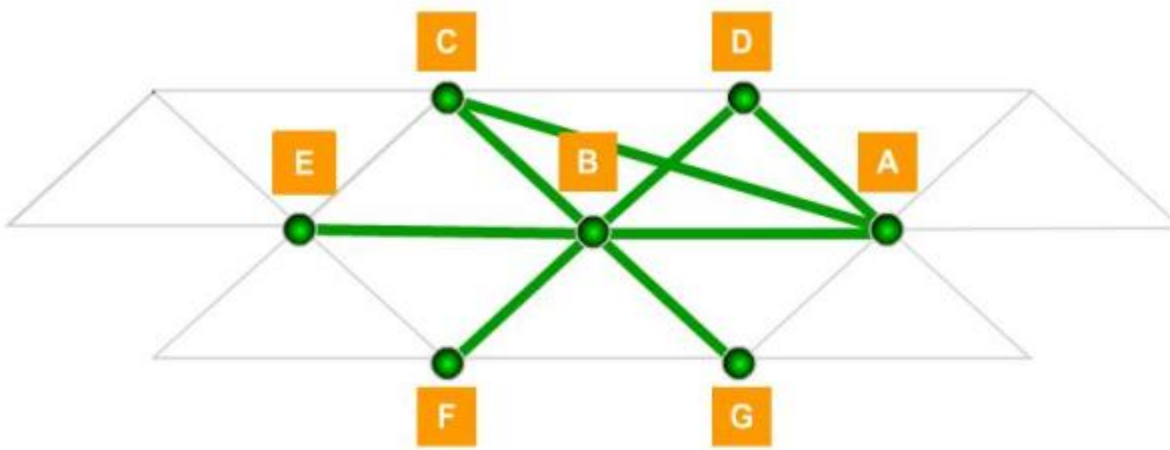
1. Calculate the number of transfers and complete the link matrix:

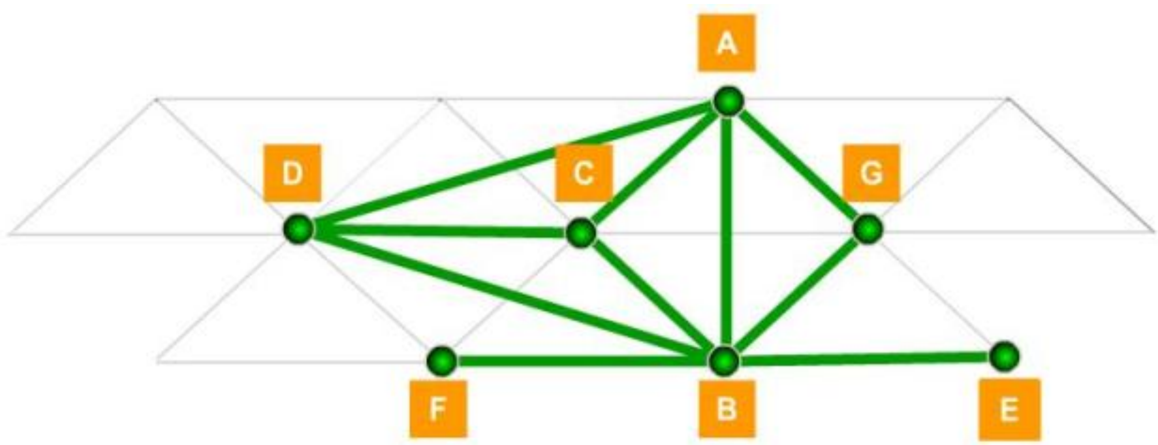
	A	B	C	D	E	F	G
G							
F							
E							
D							
C							
B							
A							

Rank the machines from the most loaded to the less loaded.

2. The link method for this situation gives several possibilities for the workshop layout. In the 3 following possibilities, which one would you choose and why?







Exercise 2:

A mechanical manufacturing shop has 7 machines (M1 to M7) and regularly produces 5 types of parts (P1 to P5).

The manufacturing routings of the mechanical parts production shop can be summarized as follows:

P1	
10	M5
20	M3
30	M2

P2	
10	M1
20	M4
30	M5
40	M7
50	M2

P3	
10	M1
20	M6
30	M3
40	M2

P4	
10	M1
20	M5
30	M3
40	M2

P5	
10	M5
20	M3
30	M7
40	M2

You also have the following information about the parts :

Parts	Quantity produced	Transfer batch
P1	90	30
P2	150	50
P3	100	50
P4	100	20
P5	300	150

Using the chain method, establish at least two layout proposals on a frame of this type: